

MANOJ SANAGAPALLI

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LinkedIn

GitHub

Portfolio

PROFESSIONAL SUMMARY

CS undergraduate specializing in AIMA at VIT-AP University, with strong foundations in Data Structures & Algorithms, OOPs, DBMS, Operating Systems, and System Design. Experienced in building full-stack and microservices-based applications with REST APIs and distributed systems. Seeking SDE internship opportunities to contribute to scalable and impactful engineering teams.

EDUCATION

- B.Tech in Computer Science and Engineering (AIMA)**, VIT-AP University, Andhra Pradesh 2024 – Present
CGPA: 8.7 / 10.0
- Higher Secondary Education**, Krishna Reddy College, Vijayawada, Andhra Pradesh 2022 – 2024
Percentage: 95.1 / 100

SKILLS

- Programming Languages:** Python, Java, C++, SQL, JavaScript
- Frontend & Backend:** React.js, Node.js, Express.js, REST APIs, Microservices
- Core CS Concepts:** Data Structures & Algorithms, OOPs, DBMS, System Design, OS, CN
- Developer Tools:** Git, GitHub, VS Code

PROJECTS

RideShield | React.js, Node.js, MongoDB, RabbitMQ, Stripe, Redis |

Jun 2026

- Engineered a parametric income-protection insurance platform for gig drivers (Uber/Rapido), automating claim payouts during rain or strike disruptions to cut processing time from days to under 30 seconds.
- Developed scalable RESTful APIs using Node.js and Express in a modular architecture, efficiently managing policy lifecycles for 10,000+ drivers.
- Built asynchronous inter-service communication using RabbitMQ, achieving a decoupled system with 99.9% message delivery reliability.
- Created a responsive React.js frontend (User & Admin Dashboards) integrated with MongoDB and Redis, implementing caching that reduced external API costs by 95% and cut latency to sub-50 ms.

GitHub Profile Analyzer | API + Data Visualization | (Live Demo)

Nov 2025

- Developed a developer analytics platform using the GitHub REST API to analyze up to 100 repositories per query, transforming raw profile data into an actionable Developer Strength Score with Chart.js visualizations.
- Engineered a side-by-side "VS" comparison engine that evaluates 10+ developer metrics simultaneously to generate a visual Developer DNA tug-of-war and compatibility score.
- Built a Vercel serverless proxy backend with secure token authentication and custom caching to increase API rate-limit capacity by 83x.

Cosmic Explorer | JavaScript, APIs, Indexed DB | (Live Demo)

Aug 2025

- Built a 3D physics-based simulation system optimizing orbital trajectory and ΔV calculations to under 50 ms latency.
- Implemented Flask REST APIs for processing 5,000+ telemetry datasets, reducing orbital computation times by 35%.
- Architected an offline-first state machine utilizing IndexedDB and local storage to cache mission diaries and telemetry streams, ensuring 100% data persistence and eliminating data loss during network dropouts.

ACHIEVEMENTS & POSITIONS OF RESPONSIBILITY

Co-Founder — (CollegePrep.Page)

Co-founded and scaled an EdTech platform serving 2600+ registered users and a 4200+ member community. Contributed to platform development, feature optimization, and user experience improvements to support scalable growth and community engagement.

- National Finalist – CTRL+SPACE Hackathon**, Newton School of Technology (Team VOID ranked Top 5 nationwide)
- Team Lead – Hackathon:** Managed frontend development and AI integration for a 4-member hackathon team, coordinating GitHub workflows and delivering the project within deadline.
- Marketing Head –** Coordinated marketing campaigns and audience engagement initiatives for **TEDxVITAP 2025**.

CERTIFICATIONS

- Introduction To Python - Microsoft
- Programming in Java - Infosys
- AI Foundations Associate – Oracle
- SQL & Relational Databases - Cognitive Class